

MACHINE LEARNING ASSIGNMENT 1

Polynomial Regression

Final solution and cross-validation results | Student BT2024146

1. Assignment and data

This work solves two separate supervised regression tasks. Each model predicts the continuous target **y** from the features supplied for its problem. The permitted total polynomial degrees are 1 to 10 for var1 and 1 to 20 for var2. Each problem has 1,000 labeled training rows and 1,000 test rows; test labels are unavailable.

Problem	Input features	Train / test	Target y: range; mean; SD	Data checks
var1	6: x1, x2, x3, x4, x5, x6 Range -1.0 to 1.0	1,000 / 1,000 rows	-9.59 to 13.95 0.76; SD 3.27	Missing 0 Train duplicates 0 Test feature duplicates 2
var2	3: x1, x2, x3 Range -1.0 to 1.0	1,000 / 1,000 rows	-29.85 to 39.75 2.24; SD 6.80	Missing 0 Train duplicates 0 Test feature duplicates 6

Training and test feature schemas match for both problems. The supplied data have no missing values; all features span -1.0 to 1.0. The sample submission contains 1,000 rows and is validated by the training script as one column named y.

2. Validation and model methodology

The 1,000 labeled rows in each problem are evaluated with shuffled RepeatedKfold cross-validation: five folds repeated twice, for 10 validation folds total (random seed 2026). Each fold uses 800 rows for training and 200 for validation. This repeated shuffled split provides several estimates of generalization while retaining the full training sample for the final fit.

PolynomialFeatures creates all monomials up to each candidate total degree. Ridge and Lasso candidates are compared across the complete permitted degree range; ordinary least-squares polynomial regression is also evaluated at degrees with fewer than 800 expanded terms. The maximum expansions contain 8,007 terms for var1 at degree 10 and 1,770 terms for var2 at degree 20, so regularization is critical for controlling complexity at higher degrees.

The feature expansion is deterministic. StandardScaler is fitted only on each fold's training partition and then applied to that fold's validation partition. Mean validation MSE and R-squared are calculated across the 10 folds. The selected degree is the lowest degree within one standard error of the minimum mean validation MSE; within that degree, the lowest-MSE candidate is selected.

Final selected models

Problem	Model	Degree	Alpha	Terms	CV MSE	CV R-squared
var1	Lasso polynomial	5	0.007	461	0.333088	0.968708
var2	Ridge polynomial	10	1	285	0.262251	0.994186

3. Degree search and model comparison

The table reports the best mean validation MSE and R-squared across candidate models at each degree. Every permitted degree was evaluated for both problems. A star marks the selected degree; the dagger marks the raw minimum for var2.

var1 degree	CV MSE	CV R-squared	var2 degree	CV MSE	CV R-squared
1	9.4013	0.1175	1	33.4131	0.2724
2	2.8267	0.7336	2	21.0831	0.5391
3	1.0010	0.9058	3	11.0045	0.7583
4	0.6694	0.9371	4	3.3031	0.9272
5 *	0.3331	0.9687	5	1.4935	0.9672
6	0.3525	0.9669	6	0.5264	0.9883
7	0.3611	0.9661	7	0.3597	0.9920
8	0.3654	0.9657	8	0.2801	0.9938
9	0.3695	0.9653	9	0.2730	0.9939
10	0.3721	0.9651	10*	0.2623	0.9942
-	-	-	11†	0.2607	0.9942
-	-	-	12	0.2619	0.9942
-	-	-	13	0.2658	0.9941
-	-	-	14	0.2745	0.9939
-	-	-	15	0.2821	0.9938
-	-	-	16	0.2862	0.9937
-	-	-	17	0.2898	0.9936
-	-	-	18	0.2953	0.9935
-	-	-	19	0.2993	0.9934
-	-	-	20	0.3067	0.9932

* Selected degree. † Lowest raw mean CV MSE for var2. Scores are means over the same 10 validation folds.

Degree 11 achieved the lowest raw cross-validation MSE. However, degree 10 was selected using the one-standard-error rule because its validation error was within one standard error of the minimum while requiring fewer polynomial terms.

For var2, degree 11 had mean CV MSE 0.260706; selected degree 10 had 0.262251, within the one-standard-error cutoff of 0.268001. Degree 10 uses 285 terms versus 363 at degree 11.

OLS benchmark

Problem	Best OLS degree	Terms	CV MSE	CV R-squared	Selected Model CV MSE
var1	4	209	0.815519	0.923266	0.333088
var2	8	164	0.284971	0.993656	0.262251

The selected regularized models outperform the best evaluated OLS polynomial for both problems. The final choices use Lasso degree 5, alpha 0.007 for var1 and Ridge degree 10, alpha 1 for var2.

Regularization rationale: Lasso vs. Ridge selection

Why Lasso for var1: The turbine system involves 6 continuous operational parameters that expand combinatorially to 461 monomials at degree 5 (and up to 8,007 terms at degree 10). In multi-variable engineering systems, physical behavior is governed by a sparse subset of interaction terms. Ridge regression (L2 penalty) forces all 461 terms to remain non-zero, accumulating estimation variance from noise monomials (best evaluated Ridge candidate in search: degree 5, alpha 10, CV MSE: 0.53845, CV R-squared: 0.9493). In contrast, Lasso regression (L1 penalty) drives non-informative terms to exactly zero, retaining only 133 active features in the final model (and averaging ~139 across CV folds) at alpha=0.007. Eliminating over 71% of the noisy terms dramatically reduces out-of-fold generalization error, achieving a 38.1% reduction in MSE (0.33309) and increasing CV R-squared to 0.9687.

Why Ridge for var2: The geothermal reservoir mapping task measures 3 spatial coordinate offsets (x1, x2, x3) representing physical sensor locations. The underlying thermal anomaly field is governed by continuous heat diffusion, where coordinates interact across dimensions in a smooth, dense polynomial manifold rather than isolated sparse terms. Expanding 3 coordinates produces 285 terms at degree 10, which exhibit severe geometric multicollinearity. Ridge's L2 penalty provides uniform, stable shrinkage of the collinear eigenvalues without discarding continuous spatial interaction terms, achieving exceptional predictive accuracy (CV MSE: 0.262251, CV R-squared: 0.9942).

4. Generalization and final model use

Training-fold and validation-fold scores for the selected models are shown below. The validation results are the model-selection estimates; no score is claimed for the hidden test labels.

Problem	Train MSE	CV MSE	Train R-squared	CV R-squared
var1	0.224879	0.333088	0.978941	0.968708
var2	0.176046	0.262251	0.996179	0.994186

The degree search shows a training-to-validation gap at higher complexity. For var1, the degree-10 candidate's training MSE was 0.200211 while its validation MSE was 0.372104; the selected degree-5 candidate had training MSE 0.224879 and validation MSE 0.333088. For var2, from degree 11 to degree 15 training MSE fell from 0.168720 to 0.151728, while validation MSE rose from 0.260706 to 0.282099. These results support using validation performance and a lower-complexity choice instead of minimizing training error.

5. Final fitting and prediction files

After model selection, each selected pipeline is refitted using all 1,000 labeled training rows for its problem. The fitted pipeline predicts the corresponding 1,000 test rows in their supplied order. The var1 and var2 models are trained and applied independently.

Prediction file	Automated checks
outputs/BT2024146_pred_var1.csv	1,000 finite numeric predictions; exactly one y column; no index column
outputs/BT2024146_pred_var2.csv	1,000 finite numeric predictions; exactly one y column; no index column

6. Reproducibility

The five instructor-provided CSV files used by this pipeline are included in data/. From the project root, install the required packages and run:

```
python -m pip install -r requirements.txt
python src/train_and_predict.py
```

The training script repeats model selection, refits both models, and writes the prediction and validation files to outputs/. Full setup instructions are in README.md.

Code: src/model_selection.py implements fold-safe polynomial model selection; src/train_and_predict.py checks inputs, refits the selected models, and writes predictions. Repository: github.com/Varun576253/MachineLearning_Assignment.